Walking in a closed disk

University of Toronto

Haocheng Sun

Fall 2024

Imagine all of you are selected to play a mathematical survival game. The settings are:

- You are at the center of a closed disk whose radius is 20m.
- You receives a finite set of vectors
$$M = \{\vec{v_1}, \cdots, \vec{v_n} : \|\vec{v_i}\| \leq 10, \sum_{i=1}^{n} \vec{v_i} = \vec{0}\}$$
- You can only walk by one vector in M each time, you can only use one vector exactly once.
- If at any step you go outside the disk, you die. If you go back to the origin safely, you survive.

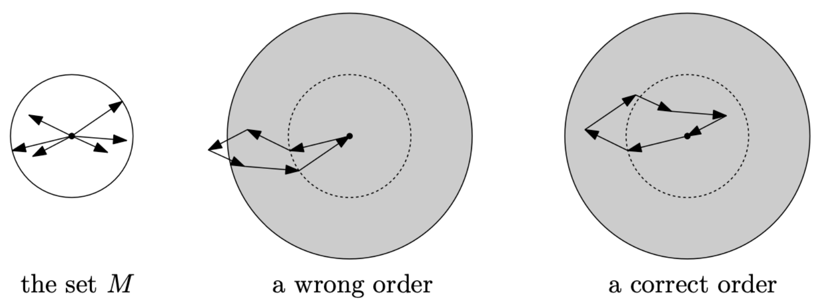

Figure 1: Example of the rules

Question: Is there a winning strategy?

Answer: Yes! For any M, we can always survive theoretically. The following theorem shows that a safe walk is possible for every finite M, and it also works for yards that are d-dimensional balls.

Theorem 0.1. *Let M be an arbitrary set of n vectors in $\mathbb{R}^d$ such that $\forall \vec{v} \in M. \|v\| \leq 1$ and $\sum_{\vec{v} \in M} \vec{v} = \vec{0}$. Then it is possible to arrange all vectors of M into a sequence $(v_1, v_2, \cdots, v_n)$ in such a way that*
$$\forall k \leq n. \|\vec{v_1} + \cdots + \vec{v_k}\| \leq d$$

We start the proof with a simple general lemma.

Lemma 0.1. *Let $A\vec{x} = \vec{b}$ be a system of m linear equations in $n \geq m$ unknowns, and let us suppose that it has a solution $\vec{x_0} \in [0,1]^n$. Then there is a solution $\vec{x'} \in [0,1]^n$ in which at least $n - m$ components are 0's or 1's.*

Proof. We will prove this lemma by induction on $k = n - m$.

- Base case: $k = 0$, in other words $n = m$.
  Just take $\vec{x'} = \vec{x_0}$, it automatically satisfies the requirement.

- Inductive step: Assume the lemma holds for $k - 1$, then prove it holds for k.
  Then the solution space has dimension at least 1, so it contains a line passing through $\vec{x_0}$. Since $\vec{x_0} \in [0,1]^n$, this line intersects the boundary of the cube $[0,1]^n$. Denote $\vec{y}$ as the intersection point. Thus $y_i \in \{0,1\}$ for some i. We now consider this y_i.
  Let us set up a new linear system with $n-1$ unknowns that is obtained from $A\vec{x} = \vec{b}$ by fixing the value $x_i = y_i$. By deleting the component y_i from the point $\vec{y} \in \partial[0,1]^n$, the new point satisfies the equation and ot is in $[0,1]^{n-1}$. By induction hypothesis, there exists a point in $[0,1]^{n-1}$ with at least $n - m - 1$ components equal to 0 or 1. By adding back the component y_i, we get a solution of the original system with at least $n - m$ components being 0 or 1

□

Definition 0.1. *Let $K = \{\vec{w_1}, \cdots, \vec{w_k}\}$ in $\mathbb{R}^d$ with $k \geq d$ and $\|\vec{w_i}\| \leq 1$. We call such K a <u>good set</u> if there exist coefficients $alpha_1, \cdots, \alpha_k$ satisfying:*

- $\alpha_i \in [0,1]$
- $\alpha_1\vec{w_1} + \cdots + \alpha_k\vec{w_k} = \vec{0}$
- $\alpha_1 + \cdots + \alpha_k = k - d$

Lemma 0.2. *If K is a good set of $k > d$ vectors, then there is some i such that $K \setminus \{\vec{w_i}\}$ is a good set of $k - 1$ vectors.*

Proof. We consider the following system of linear equations for unknowns $x_1, \ldots, x_k$:

$$x_1 w_1 + x_2 w_2 + \cdots + x_k w_k = 0 \qquad (*)$$
$$x_1 + x_2 + \cdots + x_k = k - d - 1. \qquad (**)$$

Here (*) is an equality of two d-dimensional vectors and thus it actually represents d equations. The last equation (**) is like the third condition of the definition, except that the right-hand side is $k - d - 1$; this is a preparation for showing that a suitable subset of $k - 1$ vectors in K is good.

The above system has $d+1$ equations for k unknowns. If $\alpha_1, \ldots, \alpha_k$ are the coefficients witnessing that K is good, then by setting $x_i := \frac{k-d-1}{k-d}\alpha_i$, we obtain a solution of (*), (**) lying in $[0,1]^k$.

Thus by the lemma there is also a solution $\tilde{x} \in [0,1]^k$ with at least $k - d - 1$ components equal to 0 or 1. We want to see that at least one component of $\tilde{x}$ has to be 0. Indeed, if all the $k - d - 1$ components guaranteed by the lemma happen to be 1, then all of the remaining $d + 1$ components must be 0, since all components add up to $k - d - 1$ by (**).

Now it is easy to check that for any index i with $\tilde{x}_i = 0$ the set $K \setminus \{w_i\}$ is good. Indeed, the remaining components of $\tilde{x}$ can be used in the role of the α_i in the definition of a good set. This proves the lemma. □

Lemma 0.3. *Follow the setting of K above, $\|\vec{w_1} + \cdots + \vec{w_k}\| \leq d$*

Proof.

$$\begin{aligned}
\left\|\sum_{i=1}^{k} \vec{w_i}\right\| &= \left\|\sum_{i=1}^{k} \vec{w_i} - \sum_{i=1}^{k} \alpha_i \vec{w_i}\right\| \\
&= \left\|\sum_{i=1}^{k} (1-\alpha_i)\vec{w_i}\right\| \\
&\leq \sum_{i=1}^{k} (1-\alpha_i) \|\vec{w_i}\| \\
&\leq \sum_{i=1}^{k} (1-\alpha_i) \\
&= d
\end{aligned}$$

□

Proof of the theorem. We will use induction on n.

- Base case: $n = 2$.
  This is trivial. Just notice that n cannot be one because we cannot go back to origin. And for $n = 2$, we can only have $\vec{v_1} = -\vec{v_2}$.

- Inductive step: Assume theorem holds for $n-1$, we prove it holds for n.
  We start with the set $M_n = M$, which is obviously a good set. By the second lemma, we can find a vector in M denoted as $\vec{v_n}$. Such that $M_{n-1} = M_n \setminus \{\vec{v_n}\}$ is a good set. Similarly, having constructed the good set M_k, we find can find a vector $\vec{v_k} \in M_k$ such that $M_{k-1} = M_k \setminus \{\vec{v_k}\}$ is a good set. By the third lemma, done.

□

Theorem 0.2*. *In the plane, for any such set M, there is a way to arrange the path lies in the radius $\frac{\sqrt{5}}{2}$, this is the smallest possible radius.*

Open Question: What is the smallest possible radius for the case of general d-dimension sphere?